

WTMPI Vision Camera Module

Rugged Weather-Resistant Concept — Built Around the CADDXFPV Ratel Ultra

WTMPI VISION CAMERA MODULE

RUGGED WEATHERPROOF VERSION

BUILT AROUND CADDXFPV RATEL ULTRA

A fully enclosed, adventure-ready analog camera module designed for the WTMPI platform. Built to withstand the elements and integrate seamlessly with backpack shoulder straps or external mounting points.



RATEL ULTRA KEY SPECS

- Sensor: 1/1.8"
- Resolution: 1200TVL
- FOV: 125° (4:3)
- Min. Illumination: 0.00001 Lux
- System: NTSC / PAL (Switchable)
- Input Voltage: 4.5 – 36V DC
- Dimensions: 19.5 x 19.5 x 28.1 mm
- Weight: ~13g (camera only)



BUILT FOR THE ELEMENTS

- IP66 / IP67 Rated (Splash Proof / Rain Resistant)
- Sealed Housing with O-Ring Gaskets
- UV Resistant Polycarbonate Enclosure
- Operating Temp. -20°C to 60°C
- Shock & Vibration Resistant Design

1. FULLY ENCLOSED WEATHERPROOF HOUSING



Hydrophobic Lens Window (Replaceable)

Recessed Lens Protection

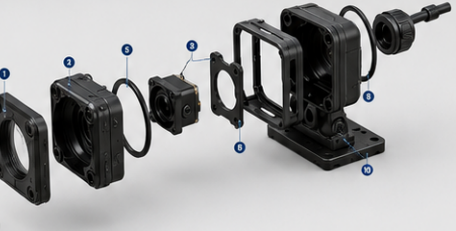
Rugged Polycarbonate Housing

316 Stainless Steel Security Screws

Sealed Cable Exit with Gland

Integrated Mounting Base (WTMPI Interface)

2. EXPLODED VIEW



- Hydrophobic Lens Window
- Front Cover (Polycarbonate)
- O-Ring Seal (Lens)
- Ratel Ultra Camera
- Silicone Vibration Dampers
- Internal Mounting Frame
- Rear Housing (Polycarbonate)
- O-Ring Seal (Rear)
- Cable Gland
- Base Mount (WTMPI Interface)

3. SHOULDER STRAP MOUNTING SYSTEM



Low Profile Mounting Base

Hook & Loop Strap Pass-Through

Tool-Less Adjustment Locking Knob

Adjustable Tilt ±30°

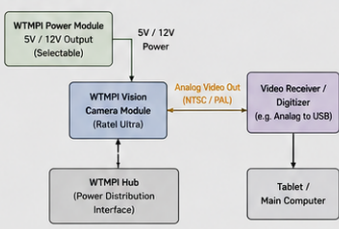
4. CABLE ROUTING



Cable routes securely down the shoulder strap using webbing loops and low-profile retention clips.

Enters WTMPI Hub inside backpack.

5. POWER / VIDEO ARCHITECTURE



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    A[WTMPI Power Module<br/>5V / 12V Output<br/>(Selectable)] -- "5V / 12V Power" --> B[WTMPI Vision Camera Module<br/>(Ratel Ultra)]
    B -- "Analog Video Out<br/>(NTSC / PAL)" --> C[Video Receiver /<br/>Digitizer<br/>(e.g. Analog to USB)]
    B --> D[WTMPI Hub<br/>(Power Distribution Interface)]
    C --> E[Tablet /<br/>Main Computer]
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6. DIMENSIONS



41mm

56mm

41mm

41mm

Weight (with mount & cable): ~85g

7. INTERFACE (WTMPI STANDARD)



WTMPI Interface Port (Power / Video)

Keyed Connector (Anti-Reverse)

Pinout (Example):

- 1 – +12V / +5V Power
- 2 – GND
- 3 – Video Out (Analog)
- 4 – ID / Module Detect

8. MOUNTING & ANGLE ADJUSTMENT



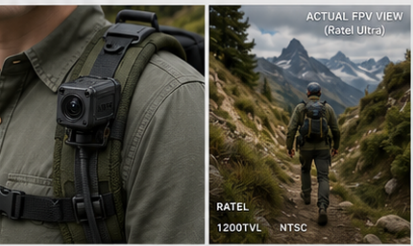
Locking knob secures desired angle. No tools required.

+30°

0°

-30°

9. IN THE FIELD



ACTUAL FPV VIEW (Ratel Ultra)

RATEL 1200TVL NTSC

10. WHAT'S INCLUDED

- WTMPI Vision Camera Module (Ratel Ultra)
- Shoulder Strap Mount with Tilt Bracket
- Hook & Loop Strap
- Cable with Sealed Gland (1m)
- Interface Dust Cap
- Quick Start Guide



11. KEY FEATURES

- Rugged & Weatherproof (IP66 / IP67)
- Adventure Ready
- Modular & Serviceable Design
- Standardized WTMPI Interface
- Power & Video over Single Cable
- Quick Mount / Quick Adjust
- Built for Exploration



WTMPI

MODULAR.
ADAPTIVE.
OPEN.

WTMPI Vision Camera Module
v1.0 Concept Design

Concept status: This document records a WTMPI design concept. Enclosure geometry, mounting dimensions, interface pinout, environmental ratings, materials, and production details shown in the artwork are **illustrative and require engineering verification** before being treated as final specifications.

WTMPI Vision — Design Intent

The WTMPI Vision module extends the WildThread Modular Pack Interface with a small external camera module. The CADDXFPV Ratel Ultra is the reference camera; the WTMPI mechanical and interface architecture is intended to remain usable with compatible future cameras.

Core Design Objectives

Objective	Concept direction
Modular mounting	Removable WTMPI-compatible base for backpack webbing and shoulder straps.
Weather resistance	Sealed enclosure concept with protected lens window, gasketed housing, and sealed cable exit.
Serviceability	Camera and enclosure components should be removable without destroying backpack integration.
Cable management	Protected routing along the shoulder strap and into the backpack's internal WTMPI hub.
Video architecture	Analog video remains separate from the ESP32-C6 controller; a suitable receiver/digitizer handles conversion where required.
Future compatibility	WTMPI Vision should accommodate alternate camera modules without redefining the entire WTMPI platform.

Proposed Signal Architecture

WTMPI Power → Vision Camera Module → analog video → compatible receiver/digitizer → tablet or main computer. The WTMPI Hub can provide protected internal routing and power distribution. Exact voltage, connector, grounding, cable, and receiver choices should be finalized after testing.

Next Engineering Tasks

1. Verify current Ratel Ultra electrical and mechanical specifications.
2. Precisely measure the camera and connectors.
3. Select the WTMPI Vision connector and safe power scheme.
4. Prototype the enclosure and adjustable shoulder-strap mount.
5. Test sealing, vibration, strain relief, thermal behavior, and field visibility.
6. Publish verified dimensions and interfaces in the WTMPI repository.

WTMPI — WildThread Modular Pack Interface

Developed by Artisans Wildthread
Open-source hardware project concept